

Sandeep Singh

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LinkedIn

GitHub

Portfolio

Kanpur, Uttar Pradesh, 273016

Profile Summary

Motivated and detail-oriented Computer Science and Engineering student with a strong foundation in Data Structures, Algorithms, Object-Oriented Programming, DBMS, Operating Systems, and Computer Networks. Proficient in C++, Python, JavaScript, and the MERN Stack (MongoDB, Express.js, React.js, and Node.js). Experienced in developing full-stack web applications and implementing RESTful APIs through academic and personal projects. Passionate about software development, problem-solving, and learning emerging technologies. Possesses strong analytical, communication, and teamwork skills, and is eager to contribute to innovative software development and technical writing projects.

Education

- Dr. A.P.J. Abdul Kalam Technical University – 7.2 CGPA** 2027
Bachelor of Technology in Computer Science and Engineering Kanpur, Uttar Pradesh
- Class XII (Intermediate) – 85.8%** 2021
UP Board Sarswati Vidya Mandir Kasia, Uttar Pradesh

Technical Skills

- Programming Languages:** C++, Python, JavaScript
- Backend Development:** Node.js, Express.js, RESTful APIs
- Frontend Development:** React.js, HTML, CSS, Tailwind CSS
- Databases:** MongoDB, MySQL
- Tools & Practices:** Git, GitHub, Postman, VS Code
- CS Fundamentals:** Data Structures & Algorithms, OOPs, OS, DBMS, CN, Software Engineering

Projects

- Hospital Management System | MongoDB, Express.js, React.js, Node.js (MERN)**  [GitHub](#)
 - Built a full-stack Hospital Management System using the MERN stack to manage patient registration, doctor appointments, medical records, and billing in real time.
 - Implemented role-based access control (Admin, Doctor, Patient) with JWT authentication and protected React routes.
 - Designed RESTful APIs with Express.js and Node.js, integrating MongoDB for scalable and persistent data storage.
 - Created a responsive React.js frontend with dynamic dashboards, appointment booking, and medical history views.
- Autonomous Car Driving System | Raspberry Pi, Python, OpenCV, Machine Learning – In Progress**
 - Building a real-world autonomous car using Raspberry Pi as the onboard controller, integrated with a camera module for live environment sensing.
 - Implementing Machine Learning models for real-time lane detection, obstacle avoidance, and steering angle prediction directly on the Raspberry Pi.
 - Using OpenCV for live video frame capture and processing to detect road lanes and obstacles under varying lighting conditions.
 - Controlling motor drivers and servo mechanisms via Raspberry Pi GPIO pins for physical steering and speed regulation of the car.

Achievements

- Built a fully functional Hospital Management System integrating patient records, appointment scheduling, and billing.
- Developing a real-world Autonomous Car Driving System using Raspberry Pi, OpenCV, and Machine Learning (In Progress).
- Selected among top participants in college-level coding competitions for algorithmic problem solving in C++.
- Consistently maintained strong academic performance throughout the 4-year B.Tech program.